

MD. MEHEDI HASAN SHOIB



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AI/ML Engineer and competitive programmer specializing in Generative AI and Agentic AI systems. Expert in retrieval-augmented generation (RAG) and multi-agent orchestration using LLM models, LangChain, LangGraph, LangServe, and LangSmith, with FAISS and Chroma vector search, REST APIs in FastAPI, Next.js and Streamlit front ends, and PostgreSQL. Additional depth in NLP, computer vision with PyTorch, and MLOps. Strong in algorithms and data structures. Seeking to contribute to production Generative AI and Agentic AI products and grow with a team shipping at scale.

WORK EXPERIENCE

Research Assistant (RA)

Health Informatics Research Lab- HIRL

📅 January 2025 – Present (Onsite)

📍 DIU, Daffodil Smart City, Birulia, Savar, Dhaka, Bangladesh

- Conducting research on medical imaging in the healthcare domain and agricultural image data in the agriculture domain.
- Writing research papers, working with computer vision, and building custom ML & Deep Learning models.

SKILLS

- **Generative AI & Agentic AI:** LLM Models, LLM Fintuning (LORA, QLORA), RAG, multi-agent orchestration, N8N, LangChain, LangGraph, LangServe, LangSmith, prompt engineering, tool calling, structured output, HITL, embedding models, semantic & hybrid search, chunking strategies, reranking, metadata filtering
- **NLP:** Tokenization, text classification, NER, sentiment analysis, text summarization, topic modeling, semantic similarity, Hugging Face Transformers
- **ML & Deep Learning:** Computer Vision, MLOps, PyTorch, TensorFlow, Supervised learning, unsupervised learning, ANN, CNN, LSTM, Transformers, Graph Neural Networks, attention mechanisms, fusion models, Explainable AI (Grad-CAM, Grad-CAM++, SHAP, LIME, TCAV), federated & split learning
- **Backend & APIs:** FastAPI, REST API design, Pydantic validation, authentication, async request handling, model-serving endpoints
- **Databases:** PostgreSQL, MySQL, Vector DB (FAISS, Chroma, Pinecone)
- **Frontend:** Next.js, React, TypeScript, Tailwind CSS, HTML/CSS, Streamlit
- **DevOps:** Docker, Git, GitHub Actions, CI/CD
- **Languages:** Python, C, C++, Java, JavaScript, SQL

PROJECTS

Enterprise Knowledge Assistant – RAG Platform for Internal Docs [🔗 \[Backend\]](#) [\[Live\]](#)

- Built an enterprise RAG assistant for internal knowledge search – document upload, automated ingestion, and a citation-aware chat workspace with confidence indicators.
- Engineered the retrieval pipeline with PDF/DOCX/TXT parsing, recursive chunking, OpenAI embeddings, and Pinecone indexing, with role-filtered retrieval so users only see authorized content.
- Implemented JWT auth with 4-role RBAC, chat history, feedback capture, audit logging, and an admin dashboard surfacing usage metrics and low-confidence queries.
- Added a RAGAS-style evaluation endpoint with LangSmith tracing to monitor answer quality; deployed via Docker Compose.
- **Tech:** FastAPI, Python, LangChain, OpenAI, Pinecone, PostgreSQL, SQLAlchemy, JWT, LangSmith, Docker, Next.js 14, React, TypeScript, Tailwind CSS

Filing Reconciler – Multi-Document Contradiction Analyst [🔗 \[Backend\]](#) [\[Live\]](#)

- Built an AI analyst platform that ingests a company's 10-K, 10-Q, earnings-call transcript, and press release to surface cross-document contradictions, score risk, and generate a cited analyst memo.
- Engineered an agentic LangGraph pipeline (ingest → parallel claim extraction → reconciliation → risk scoring → draft → self-critique) with Postgres checkpointing, two human-in-the-loop approval gates, and cost/iteration circuit breakers.
- Designed the Next.js 16 / React 19 reviewer UI – live pipeline stepper, drag-and-drop PDF upload with document tagging, and confirm/reject/edit contradiction cards with click-to-expand citations.
- Kept numeric detection fully deterministic (regex + tolerance comparison, no LLM-generated figures), reaching 1.00 precision/recall and exact-span citation match; shipped a CI-gated eval harness (F0.5, citation faithfulness, ECE) behind a provider-agnostic LLM interface.
- **Tech:** FastAPI, Python, LangGraph, Pydantic v2, Postgres, Docker, OpenAI API, Next.js 16, React 19, TypeScript, Tailwind CSS, Framer Motion

SupportLens – Customer Support Intelligence Platform [↗](#) [\[Backend\]](#) [\[Live\]](#)

- Built an end-to-end NLP platform turning raw support tickets into structured intelligence — intent/urgency classification, entity extraction, sentiment trajectories, summarization, topic discovery, semantic search, and RAG-drafted replies.
- Benchmarked classical baselines vs. fine-tuned transformers per task and deployed the winner — +0.115 macro-F1 on urgency (DeBERTa-v3) and +0.17 ROUGE-1 on summarization (FLAN-T5).
- Implemented hybrid retrieval (dense embeddings + cross-encoder rerank, hit-rate@5 0.920) with citation-grounded RAG replies behind a budget-capped LLM client (< \$0.04 total spend).
- Delivered a Next.js metrics dashboard backed by a PostgreSQL-persisted eval harness with live drift monitoring, containerized via Docker Compose with CI/CD.
- **Tech:** FastAPI, PostgreSQL, Chroma, Hugging Face Transformers, BERTopic, OpenAI, Next.js 15, TypeScript, Tailwind CSS, Docker

YouTube Transcript RAG Chatbot – Chrome Extension [↗](#) [\[Backend\]](#)

- Built an end-to-end RAG chatbot (Manifest V3 Chrome Extension + FastAPI backend) that answers questions about any YouTube video using only its transcript, deployed on Render.
- Engineered the retrieval pipeline with DOM transcript extraction, recursive chunking, OpenAI embeddings, and FAISS vector search, served via LangServe routes with a dedicated summarization path.
- Secured the shared backend with per-IP rate limiting, an LRU-capped vector store, and automatic re-indexing to survive free-tier cold starts.
- **Tech:** Python, FastAPI, LangChain, LangServe, FAISS, OpenAI API, JavaScript, Chrome Extension (MV3)

PROBLEM SOLVING

- [Beecrowd](#) [↗](#) : Top 1%, Solved: 210+
- [Codeforces](#) [↗](#) : max. newbie (rating 948)
- [LeetCode](#) [↗](#) : Solved: 20+
- [VJudge](#) [↗](#) : Solved: 50+
- **Total Solved: 700+ | Contests: 60+ (incl. onsite)**

COMPETITIONS

- **18th** – Take-Off Contest Fall 2022 Final [↗](#)
- **26th** – Unlock the Algorithm Fall-23 [↗](#)
- **45th** – Take-Off Contest Fall 2022 Slot B [↗](#)
- **42th** – Take-Off Programming Contest Fall 2022 [↗](#)

CERTIFICATIONS

- Take-Off Programming Contest Final Fall-2022 [↗](#)

EDUCATION

B.Sc in Computer Science & Engineering

Daffodil International University

 Jan 2022 – Jan 2026 (CGPA: 3.80 / 4.00)

PUBLICATIONS

- “CardioLiteNet: A Two-Stage Lightweight Autoencoder-Augmented Framework for Robust ECG Image Classification on Small Datasets,” [Online]. Available: https://doi.org/10.1007/978-3-032-15764-5_23.
- “Cauliflower leaf diseases: A computer vision dataset for smart agriculture,” [Online]. Available: <https://doi.org/10.1016/j.dib.2025.111594v>.
- “IDBGL: A unique image dataset of black gram (Vigna mungo) leaves for disease detection and classification,” [Online]. Available: <https://doi.org/10.1016/j.dib.2025.111347>.
- “SplitX-OralNet: A Privacy-Preserving and Explainable Deep Learning Framework for Multi-class Oral Disease Detection from Intraoral Images,” [Online]. Available: https://doi.org/10.1007/978-3-032-15764-5_22.
- “XAI-GIFNet: A Fusion of DenseNet121 and Efficient-NetB0 with Gradient-Based Explainability for GI Bleeding Detection from Endoscopic Imagery,” [Online]. Available: https://doi.org/10.1007/978-3-032-15764-5_8.